

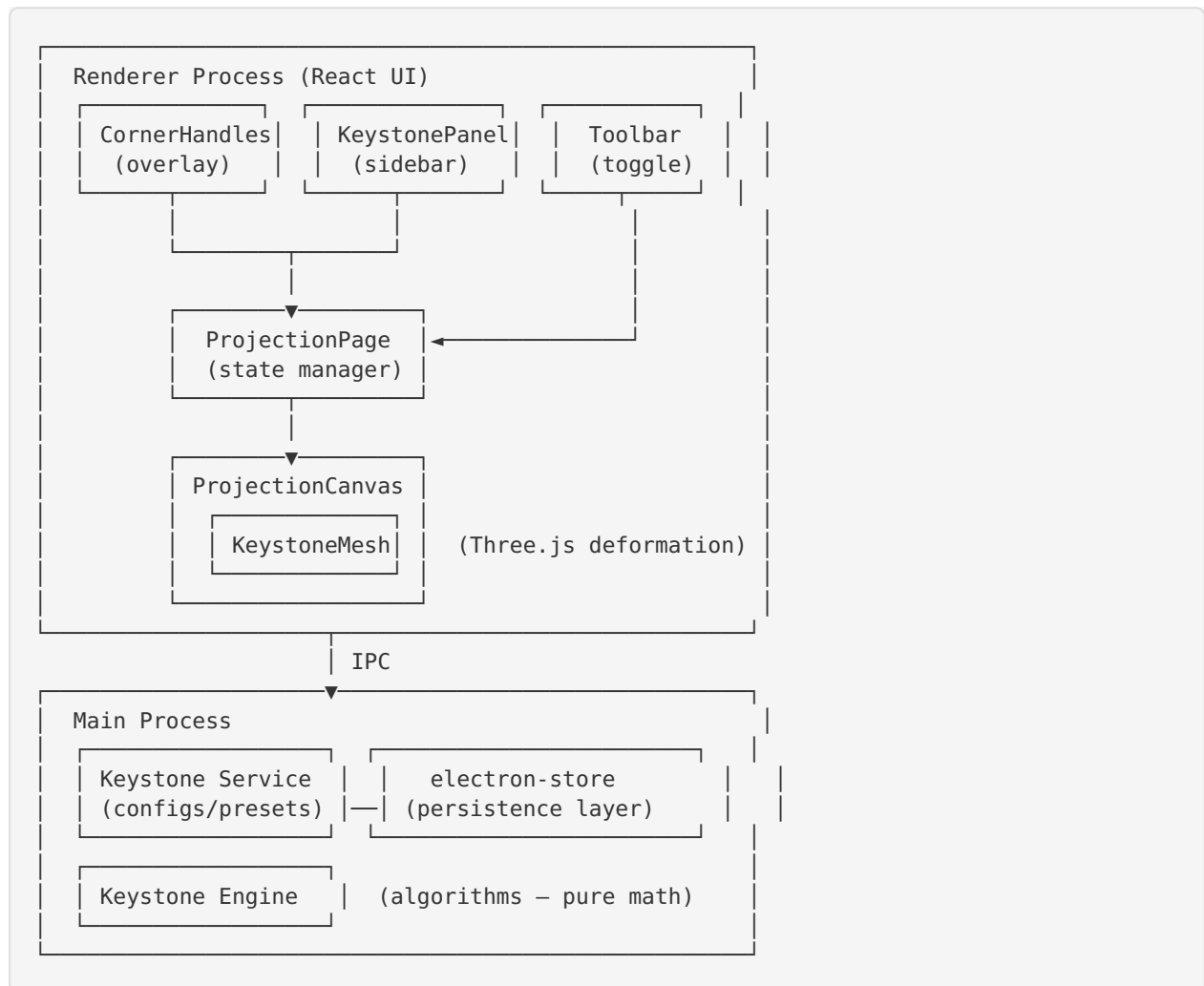
Keystone Correction Setup Guide

Overview

Keystone correction compensates for the geometric distortion that occurs when a projector is not perfectly perpendicular to the projection surface. The 4-point keystone system allows you to independently adjust each corner of the projection to achieve a perfectly rectangular image on any angled surface.

Premium Feature: Keystone correction requires a Premium or Enterprise license. See the [License](#) section for details.

Architecture



Components

Keystone Engine (`src/services/keystone-engine.ts`)

Pure mathematical algorithms for keystone correction:

- `computePerspectiveMatrix(corners)` — Computes a 3×3 homography matrix from 4 corner points using the Direct Linear Transform (DLT) algorithm
- `transformPoint(matrix, point)` — Applies a perspective transform to a 2D point
- `computeDeformedMesh(corners, subdivisions, width, height)` — Generates deformed vertex positions and UV coordinates for a subdivided plane mesh
- `toMatrix4(m3)` — Converts a 3×3 matrix to a Three.js-compatible 4×4 matrix (column-major)
- `isIdentityKeystone(corners)` — Checks if corners match the default (no correction)
- `clampCorner(point)` — Constrains a point to [0..1] range
- `snapToGrid(point, gridSize)` — Snaps a point to the nearest grid position
- `validateCorners(corners)` — Validates that corners form a non-degenerate quadrilateral

Keystone Service (`src/services/keystone-service.ts`)

State management for keystone configurations and presets:

- **Config CRUD:** `getConfig()`, `saveConfig()`, `deleteConfig()`, `resetConfig()`
- **Preset Management:** `savePreset()`, `deletePreset()`, `applyPreset()`, `getPresets()`
- **Persistence:** `loadConfigs()`, `loadPresets()` (from electron-store via IPC)

KeystoneMesh (`src/renderer/components/KeystoneMesh.tsx`)

Three.js mesh component that renders a deformed projection surface:

- Subdivided `BufferGeometry` with bilinear interpolation
- Real-time vertex updates when corners change
- Visual border (amber when deformed, indigo when identity)
- UV-correct for texture mapping

CornerHandles (`src/renderer/components/CornerHandles.tsx`)

Interactive HTML overlay for dragging corner points:

- Mouse drag with clamping to [0..1]
- Keyboard arrow keys ($\pm 0.5\%$, Shift for $\pm 0.1\%$)
- Tab to cycle through corners
- Snap-to-grid (5% increments)
- Visual feedback (colour, scale, selection indicator)

KeystonePanel (`src/renderer/components/KeystonePanel.tsx`)

Sidebar panel for keystone settings:

- Enable/disable toggle
- Numeric input for each corner (X, Y)
- Mesh quality (subdivision) slider
- Snap-to-grid toggle
- Preset save/load/delete
- Reset to default

- Keyboard shortcut hints

Setup Steps

1. Enable Keystone Correction

1. Open the Projection Mapper application
2. In the left sidebar, find the **Keystone Correction** section
3. Check the **Enabled** checkbox
4. The projection surface will now respond to corner adjustments

2. Adjust Corners

Method A: Interactive Drag

1. Click the  **Keystone** button in the toolbar (or click **Edit Corners** in the sidebar)
2. Drag any of the 4 corner handles (TL, TR, BR, BL) to adjust
3. The projection surface deforms in real-time

Method B: Numeric Input

1. In the Keystone Correction panel, enter precise X and Y values for each corner
2. Values range from 0.000 to 1.000 (normalised coordinates)

Method C: Keyboard

1. Click on a corner handle to select it
2. Use **Arrow keys** to move (0.5% per press)
3. Hold **Shift** for fine adjustment (0.1% per press)
4. Press **Tab** to cycle through corners
5. Press **Escape** to deselect

3. Save a Preset

1. In the **Presets** section of the Keystone panel, enter a name
2. Click **Save** or press **Enter**
3. The current corner positions are saved as a preset

4. Load a Preset

1. Click on a preset name in the list
2. The corners are immediately applied

Configuration

KeystoneConfig Schema

```
interface KeystoneConfig {
  id: string;
  projectorId: string;
  name: string;
  corners: [Point2D, Point2D, Point2D, Point2D]; // [TL, TR, BR, BL]
  meshSubdivisions: number; // 1-32 (default: 8)
  enabled: boolean;
  createdAt: string;
  updatedAt: string;
}
```

Corner Coordinate System

- **Normalised range:** `[0..1]` for both X and Y
- **Origin:** Bottom-left = `(0, 0)`
- **Top-right:** `(1, 1)`
- **Default corners** (no correction):
 - Top-Left: `(0, 1)`
 - Top-Right: `(1, 1)`
 - Bottom-Right: `(1, 0)`
 - Bottom-Left: `(0, 0)`

Mesh Subdivisions

Higher subdivision values produce smoother deformation at the cost of more geometry:

Subdivisions	Vertices	Triangles	Use Case
1	4	2	Minimal correction
4	25	32	Light keystoneing
8	81	128	Standard (default)
16	289	512	Heavy correction
32	1089	2048	Maximum quality

IPC Channels

Channel	Direction	Description
<code>keystone:getConfig</code>	Renderer → Main	Get keystone config for a projector
<code>keystone:saveConfig</code>	Renderer → Main	Save/update keystone config
<code>keystone:deleteConfig</code>	Renderer → Main	Delete keystone config
<code>keystone:getPresets</code>	Renderer → Main	Get presets for a projector
<code>keystone:savePreset</code>	Renderer → Main	Save a new preset
<code>keystone:deletePreset</code>	Renderer → Main	Delete a preset
<code>keystone:reset</code>	Renderer → Main	Reset to default corners

Feature Gating

Keystone correction is a **Premium** feature. The `keystone_correction` flag must be present in the user's enabled features to use this functionality.

When the feature is not enabled:

- The Keystone panel shows a “PREMIUM” badge
- A description of the feature is displayed
- An “Upgrade to Premium” button is shown

Persistence

Keystone configurations and presets are persisted via `electron-store` :

- **Configs:** `keystone.configs` key — Array of `KeystoneConfig` objects
- **Presets:** `keystone.presets` key — Array of `KeystonePreset` objects

Data is loaded on application startup and saved automatically when changes are made through the UI.

Troubleshooting

Projection surface doesn't deform

- Ensure keystone correction is **enabled** in the sidebar panel
- Check that you have a Premium license with the `keystone_correction` feature
- Verify that corners have been moved from default positions

Corner handles not visible

- Click the  **Keystone** button in the toolbar to enter edit mode
- Ensure keystone correction is enabled

Deformation looks incorrect

- Reset to default (click  **Reset to Default**)
- Increase mesh subdivisions for heavy corrections
- Ensure corners form a valid (non-self-intersecting) quadrilateral

Preset not loading

- Check that the preset was saved for the correct projector
- Maximum of 20 presets per projector